

Infant Morality Study

Malec dotplot description/conclusion

One proportion test

p =population proportion of babies who choose the “helper” object

H_0 : Babies are equally likely to choose the “helper” or “hinderer” object

In notation: $p=0.5$

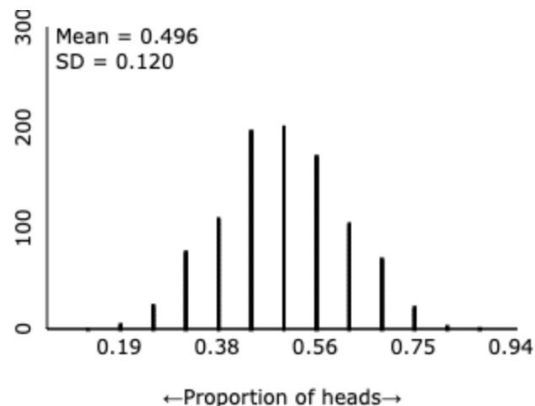
H_a : Babies are more likely to choose the “helper” object than the “hinderer” object

In notation: $p>0.5$

Malec - description of dotplot / study conclusion

This dotplot displays the results of 1000 coin-flipping simulations where each dot represents the proportion of heads that occurred in 16 flips of a coin. The proportion of heads achieved is bell-shaped and varies from about .125 to .875 with a typical proportion of heads being about .5, give or take .12. Most of the coin-flipping simulations resulted in between .38 and .66 of the flips landing on heads, and almost all resulted in between .25 and .75 landing on heads.

This coin-flipping simulation models what would happen if the babies in the study were selecting the helper/hinderer at random. Under this assumption, we see that it would be extremely unlikely for the babies to select 14 or more helper toys, since $14/16=0.875$: landing on heads 14 or more times occurred only twice in 1000 simulations, making the probability of this occurrence (the p-value) equal to $2/1000 = 0.002$. This probability is so small, it is not plausible that the babies were selecting the toys at random.



☒ Summary Statistics

